POLICY-IMPACT · DESIGN PROPOSAL · NO-HERMES OPTION

One assistant for policies, documents and spreadsheets

20 October 2026 · Draft for discussion · An alternative to the Hermes-based policy-impact build

The scope grows from comparing policies to helping staff **write policies, build documents and spreadsheets, and edit them** in the tools they open every day. For that scope we propose dropping Hermes and running **one assistant of our own, built with LangGraph, that is also the Overwatch**. Staff work in a chat window and in a document editor, and every change the assistant suggests is shown first and applied only when a person accepts it.

Decisions in this proposal

- **One agent, not two layers.** Our own LangGraph graph understands the request, calls tools, checks the result and asks for approval. The tool gate and answer checks are part of it, not a guard around someone else's agent.
- **The model suggests; code makes the change; a person accepts.** The v0.1 rule that code builds every answer now covers editing too. The model never writes a file directly.
- **A document editor next to the chat.** Staff open, view and edit Word and Excel files in a self-hosted browser editor, with an optional assistant pane inside desktop Word and Excel later.
- **Use cases are packs.** Policy impact, policy writing, documents, spreadsheets and each outside tool are separate packs with their own tools, checks and approval rules.
- **Everything stays on the sealed network** unless the client decides that Microsoft 365 is allowed.

Why drop Hermes for this scope

- **The control is the real work.** The tool gate, checks, approvals and audit would be built anyway. Without Hermes they become the assistant instead of a fence around a large, weekly-released codebase that is not being reviewed.
- **Knowing who is asking is built in.** No signed-token workaround is needed to tell tool calls which person and chat they belong to.
- **Outside tools become safe to add.** Each connector (an MCP server or a small adapter) is a tool behind our own gate, switched on per pack and per user.
- **Less to lock down.** No Hermes settings list, no read-only config checks, no stray outbound calls to switch off.

> **What we give up.** Hermes provides a ready-made agent loop, chat-title and housekeeping tasks, and an OpenAI-compatible API. A LangGraph agent loop is a small amount of code, and the OpenAI-compatible endpoint is a thin FastAPI layer, so the cost is a few days of build, paid back in a smaller system we fully understand.

System architecture

Everything below the proxy runs on the client's GPU server under Docker Compose, on internal networks with no route to the internet. Staff reach three front ends through the proxy; all three talk to the same assistant.

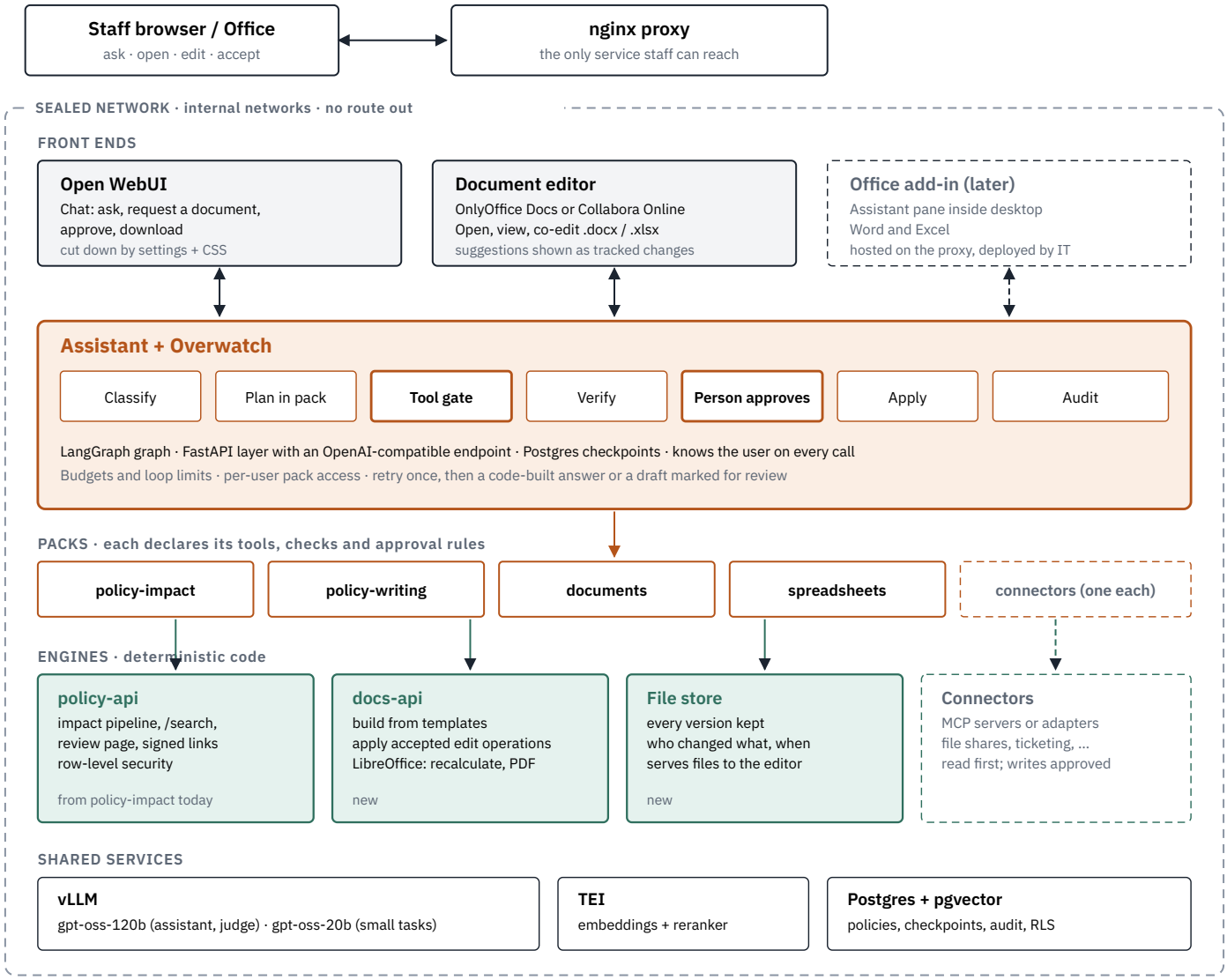


Figure 1. Three front ends, one assistant. The assistant reaches engines and connectors only through its own tool gate, and nothing is changed until a person approves it. Dashed boxes come after the pilot.

Assistant, Overwatch and packs · Deterministic engines · Front ends · Shared infrastructure

What staff can do, and how each stays safe

Job	What happens	What keeps it right
Compare policies	The fixed impact pipeline runs; code builds the answer, table and workbook.	Unchanged from today: repeatable verdicts, one audit row per model decision.
Write a policy	The assistant drafts from the policy template, section by section, quoting existing policies as sources.	The draft goes through the **impact check** before it can be published, so conflicts show first. Publishing needs a named approver and creates a new version.
Build a document	The model writes a structured outline; docs-api renders it from an approved template (Word, PDF, PowerPoint).	Facts, names, dates and figures must come from sources or from the person; branding comes from the template.
Build a spreadsheet	The model supplies the data and layout; **code writes the formulas**.	LibreOffice recalculates the workbook and the totals are checked before anyone sees it.
Edit an open file	The model suggests edit operations ("replace clause 4.2 with…", "set B7 to =SUM(B2:B6)").	Shown as tracked changes or highlighted cells. The person clicks **Accept** or **Undo**; every version is kept.
Use another tool	A connector reads from or writes to an outside system.	Switched on per pack and per user. Reading first; anything that writes or sends needs approval.

How an edit works

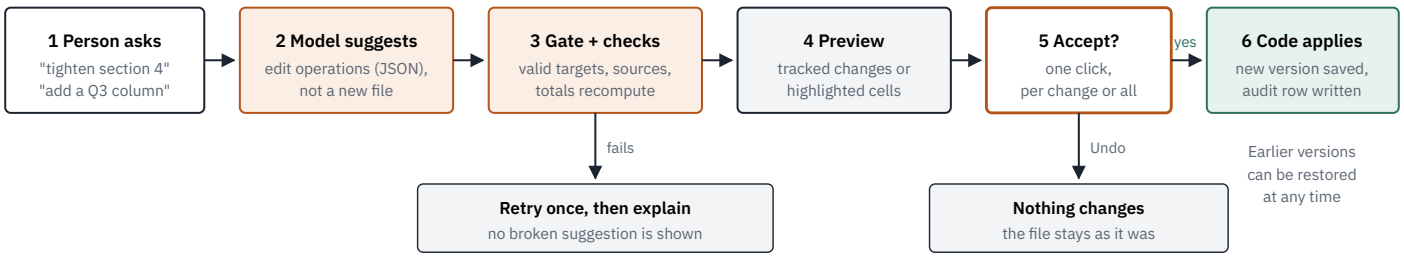


Figure 2. The model never writes the file. It suggests operations, code checks and shows them, a person accepts, and code applies them as a new version.

Where staff edit: three options

This is the decision that shapes the front end. The assistant, packs and engines are the same in all three; only where the file opens changes.

Option	What staff see	Strengths	Watch out for
1. Browser editor on our server (OnlyOffice Docs or Collabora Online) *Recommended for the pilot*	Word and Excel files open in a browser tab next to the chat	Fully inside the sealed network; co-editing; APIs to show and apply changes; OnlyOffice's AI plugin can point at our assistant	OnlyOffice Community is free (AGPL plus attribution terms) and no longer capped at 20 connections since 9.4; Collabora's free CODE edition is not for production, which needs a subscription
2. Office add-in in desktop Word and Excel	An assistant pane inside the Word or Excel they already use	No new editor to learn; works on the internal network when the add-in is hosted on our proxy	Needs an internal HTTPS certificate and deployment by the client's IT; more front-end work; depends on Office versions in use
3. Microsoft 365 online (Excel Online, SharePoint, OneDrive)	The familiar Microsoft web apps	Most familiar; files already live there for many clients	Files and Microsoft Graph calls leave the sealed network. This is a policy decision for the client, not a technical one

> **Recommendation.** Start with option 1 for the pilot, because it keeps the sealed network intact and needs no help from the client's IT. Add option 2 for staff who live in desktop Excel. Consider option 3 only if the client confirms that their data may be held in Microsoft 365.

Packs

Pack	Tools (examples)	Checks	Needs approval
policy-impact	search, find policy, run check, get result, make files	citations exist; claims match verdicts	no (read-only)
policy-writing	search, draft section, run impact check on draft, submit for approval	every rule cites a source or is marked new; impact check run on the final draft	publishing a new version
documents	make document, suggest edits, export PDF	no invented names, dates or amounts; template used	applying edits
spreadsheets	make workbook, suggest cell and formula edits, recalculate	totals recompute; formulas written by code; no hard-coded results	applying edits
connectors	one per outside tool, read first	allowed per user; results cited	any write or send

Research: document editors (October 2026)

We compared the self-hosted editors that could sit next to the chat. Versions and licence facts were checked on 20 October 2026; prices and anything marked *unverified* must be confirmed with the vendor before purchase.

Editor	Licence and cost	Word and Excel fidelity	Hooks for the assistant	Verdict
ONLYOFFICE Docs 10.0.0, 12 Oct 2026	Community: free, AGPL v3 plus attribution and labelling terms; 20-connection cap removed in 9.4. Enterprise or Developer: annual licence for support and extras	Built on Microsoft's own file formats (.docx, .xlsx) natively; the strongest of the options	Document Builder, plugins and connector APIs; AI plugin with custom assistants and local models	**Recommended** for the pilot
Collabora Online 26.04 (Oct 2026)	CODE edition free but not for production; production needs a subscription (price unverified)	LibreOffice engine: excellent for ODF, good but less exact for .docx and .xlsx; Excel formula fidelity unverified	WOPI, Python (UNO) scripting; 26.04 added a "manage tracked changes only" mode	Fallback if OnlyOffice fails a spike
Euro-Office	Open-source fork of ONLYOFFICE (Nextcloud, IONOS, Proton), March 2026	Same engine as ONLYOFFICE	Unverified	Watch only: 1.0 not confirmed and a dispute with ONLYOFFICE's owner is open
Univer	Apache-2.0 core; collaboration and .xlsx import/export need a paid Pro key	Spreadsheets only; no Word-style tracked changes	Strong programming model, but we would build the editor screens	Only for a custom spreadsheet screen later
Microsoft Office Online Server	Retired; support ends 31 Dec 2026	—	—	Do not use

Recommendation

- **ONLYOFFICE Docs Community 10.0**, self-hosted on the sealed network, for the pilot. Free, unlimited connections, closest to Word and Excel.
- Keep **our own docs-api** as the way edits are applied, so the assistant does not depend on the editor's AI plugin.
- Move to ONLYOFFICE Enterprise only if the client wants vendor support or features the free edition lacks.
- **Collabora** is the fallback if a spike below fails.

Rest of the stack, checked

- **LangGraph 1.2.16** (16 Oct 2026), MIT licence; Postgres checkpointer 3.2.0. Tracing is off unless `LANGSMITH_TRACING=true` is set.
- **Open WebUI 0.12.3** takes any OpenAI-compatible endpoint. Keeping its branding avoids the licence's 50-user limit on rebranding.
- **vLLM 0.32** runs gpt-oss-120b tool calls with `--tool-call-parser openai --enable-auto-tool-choice`. A reported parser issue (#58825) can turn stray text into a fake tool call; the gate must reject unknown tool names.
- **Office add-in:** Word's tracked-changes API exists, but offline loading of office.js and IT deployment paths are unverified. Kept for after the pilot.

R&D: test the hardest parts first

Five short experiments run before the main build. Each has a pass mark set now, and a fixed fallback if it fails, so the plan changes on evidence rather than opinion.

#	Experiment	Passes when	If it fails	Time
1	**Tracked changes round trip.** The assistant suggests edits to a real policy .docx; they show in ONLYOFFICE as tracked changes, are accepted there, and still look right in desktop Word	10 of 10 test edits show correctly in both editors, with the right author and no lost formatting	docs-api writes the tracked changes into the file itself, and the editor reloads it; if that also fails, try Collabora	2 days
2	**Excel fidelity.** Open and save 20 real client workbooks in ONLYOFFICE	All formulas, named ranges, conditional formatting and pivot tables survive; recalculated totals match Excel	Edit in desktop Excel through the add-in for workbooks that fail; record which features break	2 days
3	**Sealed network.** Run ONLYOFFICE, Open WebUI and the assistant with all outbound traffic blocked and logged	Zero outbound connection attempts over a day of use, including start-up and licence checks	Block at the firewall and document each attempt; replace any part that will not work without the internet	1 day
4	**Model reliability.** 30 editing and drafting tasks through the assistant on gpt-oss-120b	At least 90% produce valid edit operations first time; 100% caught by the gate when invalid	Narrow each task to fewer tool calls, add worked examples, or try LangChain's `create_agent` against a hand-written graph	3 days
5	**Policy writing end to end.** Draft one real policy update, run the impact check on it, approve and publish	A policy owner says the draft saved them time and the impact check found the known conflicts	Keep the assistant to section-by-section suggestions only, without whole-policy drafts	2 days

> **About two weeks of R&D.** Spikes 1 to 3 decide the editor; spike 4 decides how much the assistant does on its own; spike 5 checks that the whole idea is worth building. If spikes 1 and 2 both fail for ONLYOFFICE and Collabora, the editor moves to the Office add-in and the pilot waits for the client's IT.

Sources

- ONLYOFFICE DocumentServer releases (github.com/ONLYOFFICE/DocumentServer/releases): v10.0.0, 12 Oct 2026; v9.4.0, 19 May 2026. Connection cap removal: linuxiac.com, 20 May 2026. AI plugin: onlyoffice.com blog (May 2025, Jan 2026).
- Collabora Online CODE 26.04 release notes and 26.04 release blog (collaboraonline.com); The Register, 25 Jun 2026 (CODE not for production) and 2 Apr 2026 (Euro-Office).
- Univer (github.com/dream-num/univer). Office Online Server retirement: petri.com.
- PyPI: langgraph 1.2.16, langgraph-checkpoint-postgres 3.2.0, open-webui 0.12.3 (licence clause 4), vllm 0.32.0. docs.vllm.ai tool calling; vLLM issue #58825; LangGraph observability docs; Microsoft Word API requirement sets.
- Unverified: Collabora subscription price; Euro-Office 1.0; ONLYOFFICE AI plugin writing tracked changes; office.js offline hosting; Excel API requirement sets; gpt-oss multi-step reliability.

What carries over from policy-impact

- **Keep:** the impact pipeline, Postgres with row-level security, the review page, signed downloads, the evaluation set, the containment scripts and internal networks.
- **Becomes part of the assistant:** the model gate (approved prompt, temperature 0, one tool call at a time) and the user scope from Open WebUI, now passed directly instead of through a signed token in the chat.
- **Removed:** the Hermes container, its plugin, its read-only configuration and its checksum service.

Build order

#	Step	Passes when
0	R&D spikes 1 to 5 (about two weeks)	Each meets its pass mark, or its fallback is chosen
1	Assistant core: LangGraph graph, OpenAI-compatible endpoint, gate, audit; Open WebUI pointed at it	Chat works with the internet cut; the gate refuses unknown, out-of-order and over-budget calls
2	policy-impact pack on the new core	The evaluation cases give the same verdicts as the Hermes build
3	File store and document editor (option 1)	A staff member opens, edits and saves a Word and an Excel file; every version is kept
4	documents and spreadsheets packs, with suggested edits	Suggestions show as tracked changes or highlighted cells; Accept applies them, Undo leaves the file unchanged; totals recompute
5	policy-writing pack	A draft policy runs through the impact check and can only be published by a named approver
6	Pilot	All tests below pass
7	Later: Office add-in, then connectors one by one	Each passes its own pack evaluation before it is switched on

Tests before the pilot

1. **Same verdicts:** impact results match the current build on the evaluation cases.
2. **No silent edits:** no file changes without an Accept, and Undo always leaves the file as it was.
3. **Numbers:** every total in a generated or edited workbook recomputes to the same value; no formula results typed in by the model.
4. **Sources:** every policy id and section cited exists; documents contain no names, dates or amounts that were not supplied.
5. **Access:** a user cannot see or edit another unit's policies or files (row-level security and file store both checked).
6. **Containment:** no connection leaves the server from any container, the editor included.
7. **Runaway:** a request built to loop stops at the budget with a clear message.
8. **No-help test:** a staff member writes a short policy, builds a workbook and accepts an edit without help.

Check first, and open questions

- **Where do staff edit today:** desktop Word and Excel, Microsoft 365 online, or both? This picks the front end.
- **Can data live in Microsoft 365?** If not, option 3 is off the table.
- **Editor licence:** confirm OnlyOffice's Community terms (attribution and labelling of modified versions) with counsel, or budget for its Enterprise edition if support is needed.
- **LangGraph agent helpers:** LangChain 1.x presents `create_agent` with approval middleware; whether to use it or a hand-written graph is decided in spike 4.
- **Model reliability:** measured in spike 4 before the main build.
- **Which outside tools first:** name the first one or two connectors so their packs can be designed.